

3(a)(i)	D ; B ;	2
3(a)(ii)	any four from: 1 thylakoid stacks or discs / many thylakoids, for, increased or large, surface area ; 2 increased or large, surface area maximises the absorption of light (energy) / AW ; Ignore trap (light energy) <i>mp 3–5 thylakoid membranes / grana contain</i> 3 photosynthetic / named / primary / accessory, pigments to absorb light (energy) ; 4 electron carriers / ETC, to, transfer / release, energy (from excited electrons) ; 5 photosystems / antenna complex and reaction centre, are light harvesting structures ; 6 thylakoid space or lumen to, form proton gradient / have high concentration of protons ; 7 thylakoid membrane is (relatively) impermeable, to maintain the proton gradient (for chemiosmosis) ; 8 ATP synthase to make ATP ; 9 contains oxygen-evolving complex / OEC, for the photolysis of water ;	4

Question	Answer	Marks
3(b)(i)	any three from: 1 the pH is higher in the light than in the dark ora or the pH increases (from dark) to light or the pH decreases (from light) to dark ; 2 the pH, increases sharply / AW, when changed to light ; 3 the pH, levels off / AW, in light ; 4 the pH decreases, gradually / less steeply, when returned to the dark or decrease in pH does not return to original (dark) pH ; 5 <i>ref. to</i> fluctuations anywhere / described ; 6 comparative figures to support ; (mp1) dark(1) ~ pH 7.31–7.6 vs light ~ 7.52–7.61 vs dark (2) ~ 7.51–7.38 (mp2) from pH 7.36–7.56 in 36 s (1½ squares and each square 24 s) (mp4) from pH 7.56–7.4 in 288 s (12 squares)	3
3(b)(ii)	any three from: 1 the stroma has higher pH when H⁺ ions, move, out (of stroma) / into thylakoid, space / lumen ; 2 (leads to) increased H⁺ concentration in the thylakoid, space / lumen ; 3 (then) H⁺ diffuse back (into stroma) through ATP synthase ; 4 AVP ; e.g. idea not to support theory	3